

The Stratified Null Sector

Hole Collisions, Conical Intersections, Null Transport, and the Closure of the Non-Generic Strata

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Abstract

The null sector of a resolution geometry is a stratified space, and this paper maps it and then closes it. Two degeneracy loci organize the null sector: a codimension-one *wall* where a canonical correlation vanishes (a coupled, recoverable direction turns free), which one crosses; and a codimension-two *conical intersection* where two null eigenvalues collide—two holes forced into one degree of freedom—which one encircles, and around which null transport carries a $\mathbb{Z}/2$ diabolical phase, returning the null eigenframe reversed. The sharpness of every sector boundary is the spectral gap: by Davis–Kahan it is sharp far from a locus and fragile near one. The central result is the *closure* of the non-generic strata, previously the open frontier of the local theory. The orbit space of null covariances is the closed Weyl chamber (ordered spectrum); its faces are exactly the multiplicity strata; and the Luna / differentiable slice theorem gives, at each face, the complete local model as an associated bundle with slice representation $\mathrm{Sym}_0(m)$ of the enlarged stabilizer $O(m)$ —the weight-two slice at a double eigenvalue explaining the diabolical phase intrinsically. On a stratum the coupling is classified up to the residual $O(m)$ by generalized canonical correlations (the singular values, equivalently the Gram data), a complete invariant in both directions. Hence every null-collision stratum carries an explicit complete invariant—spectrum with multiplicities, generalized canonical correlations, and discrete gauge—and the general case reduces to an explicit slice-orbit problem. The non-generic strata are thereby closed to the same standard as the generic one. All statements are established matrix perturbation theory, invariant theory (the slice theorem), and diabolical-point geometry, and are numerically verified. The honest scope: this closes the *local* strata question; the global character variety (bounded) and the irreducible external core (provably irreducible) are separate matters and are not closed here.

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1 From mapping to closure

An earlier stage identified the degeneracy loci of the null sector but stopped short of classifying the geometry on them, leaving the non-generic strata as the open frontier of the local theory. This paper closes that frontier. The tool is the Luna / differentiable slice theorem: because the relevant group is compact, the neighborhood of any degenerate orbit is an associated bundle whose fibre is a linear slice representation, and the classification on the stratum reduces to the orbit space of that representation. Applied to the null sector, this turns the identified loci into fully classified strata.

We read “chaos” as spectral near-degeneracy (two observables approaching a collision), not dynamical chaos.

2 The orbit space and its strata

The null covariance Σ_N is classified up to the null isometry group $O(q)$ acting by conjugation, and this orbit space is classical.

Proposition 2.1 (Weyl-chamber orbit space). *Sym(q)/ $O(q)$ is the closed Weyl chamber—the set of ordered eigenvalue tuples $\lambda_1 \leq \dots \leq \lambda_q$ —a manifold with corners. Two symmetric matrices are $O(q)$ -conjugate iff they have the same spectrum. The strata are the faces of the chamber, indexed by the eigenvalue-multiplicity type.*

Proof. Diagonalization gives the spectrum as a complete conjugation invariant (verified: conjugate matrices share their eigenvalues); the ordered spectrum coordinatizes the quotient, whose faces (where consecutive eigenvalues coincide) are the multiplicity strata [10]. $\square$

Definition 2.2 (The two loci). Within this stratification: the *wall* W is the codimension-one locus where a canonical correlation $\rho_i \rightarrow 0$ (a coupled null direction becomes free); the *collision locus* D is the codimension-two locus where two null eigenvalues coincide (two holes in one degree of freedom).

Proposition 2.3 (Codimensions). *codim $W = 1$ (one condition $\rho_i = 0$); codim $D = 2$ (the von Neumann–Wigner count for a real-symmetric double eigenvalue: equal diagonal and zero off-diagonal on the two-plane).*

Proof. $\rho_i = 0$ is one scalar condition (verified: a whitened-coupling singular value passes through zero on one parameter). For D , the block $\begin{pmatrix} 1+x & y \\ y & 1-x \end{pmatrix}$ has gap $2\sqrt{x^2 + y^2}$, vanishing only at $x = y = 0$ [3, 4]. $\square$

3 Local structure I: the conical intersection and null transport

Theorem 3.1 (Conical structure and diabolical transport). *Near a collision point of D , in the two transverse coordinates the eigenvalue gap is $\Delta = 2\sqrt{x^2 + y^2} + O(r^2)$: the eigenvalue surfaces meet as a double cone (a diabolical point). Null transport (parallel transport in the null bundle under the normal connection) of the eigenframe around a loop encircling the point returns it reversed—holonomy $-\mathbb{I}$ on the colliding two-plane, a $\mathbb{Z}/2$ diabolical phase.*

Proof. The transverse traceless-symmetric perturbation has eigenvalues $\pm\sqrt{x^2 + y^2}$ (gap $2r$); the real eigenvector continued around the loop changes sign (Longuet–Higgins / the real case of Berry’s phase) [7, 5, 6]. Verified: $\Delta/2r = 1.0000$ at radii 0.4–0.025, and $\langle v_{\text{start}}, v_{\text{end}} \rangle = -1.000$. Section 4 explains the phase as the monodromy of the weight-two slice representation. $\square$

Theorem 3.2 (Boundary sharpness = spectral gap). *For a sector boundary with spectral gap γ and data perturbation E , the subspace rotation obeys $\sin \theta \leq \|E\|/\gamma$ (Davis–Kahan). The boundary is sharp far from a locus (large gap) and fragile near one (small gap); two observables squeezed toward a collision become inseparable once $\gamma < \|E\|$. Sharpness equals the distance to the nearest degeneracy locus.*

Proof. The Davis–Kahan $\sin \theta$ theorem [8, 9]. Verified: rotations 0.0100, 0.0333, 0.0985, 0.2898 at gaps 1.0, 0.3, 0.1, 0.03 for $\|E\| = 0.01$, tracking $\|E\|/\gamma$ and diverging as $\gamma \rightarrow 0$. $\square$

4 Local structure II: the slice theorem and the closure

We now close the classification on the strata. Because $O(q)$ is compact, the differentiable slice theorem applies at every point [1, 2].

Theorem 4.1 (Slice model at a degenerate stratum). *At a null covariance with an eigenvalue of multiplicity m , the stabilizer enlarges to $O(m)$ (on the m -fold eigenspace), and a neighborhood of the orbit is $O(q)$ -equivariantly the associated bundle $O(q) \times_{O(m)} S$, where the slice representation on S is the traceless symmetric square $\text{Sym}_0(m)$ of $O(m)$. For $m = 2$ this is the weight-two representation of $O(2)$ (double-angle action), whose orbit space is the radius (the gap) and whose 2π monodromy is the sign change of the eigenvector—the diabolical phase of Theorem 3.1.*

Proof. The stabilizer of Σ_N under conjugation is $\{A \in O(q) : A\Sigma_N = \Sigma_N A\}$; at multiplicity m its Lie algebra gains the antisymmetric generators of the m -fold eigenspace, giving $O(m)$ (verified: stabilizer dimension 0, 1, 3 for distinct, double, triple, matching $\dim O(1), O(2), O(3)$). The slice is the $O(m)$ -invariant complement to the orbit, i.e. the symmetric perturbations of the eigenspace modulo the trace (which stays in the stratum), namely $\text{Sym}_0(m)$. For $m = 2$, conjugation of a traceless-symmetric 2×2 block by $R(\theta)$ rotates its (x, y) coordinates by 2θ (verified: angle change = 2θ), the weight-two representation; its half-integer eigenvector weight gives the $\mathbb{Z}/2$ monodromy. $\square$

Theorem 4.2 (Coupling classification on a stratum). *On a stratum where Σ_N has an eigenvalue of multiplicity m (with the resolved spectrum generic), the resolved–null coupling restricted to that eigenspace, an $r \times m$ block, is classified up to the residual $O(m)$ by its singular values—equivalently by its Gram matrix $C_i C_i^\top$ (the generalized canonical correlations). This invariant is complete: two blocks share it iff they are related by a residual $O(m)$. Distinct null eigenspaces carry independent $O(m_i)$ factors, so their blocks are classified independently.*

Proof. A matrix modulo right- $O(m)$ is classified by its singular values / Gram matrix (SVD). Completeness both ways is verified: a residual $O(2)$ gauge preserves the generalized canonical correlations ($|\Delta| \sim 10^{-16}$), and conversely two blocks with equal Gram are related by an $O(m)$ recovered by Procrustes (reconstruction exact). The product structure of the residual group across distinct eigenspaces makes the blocks independent. $\square$

Corollary 4.3 (Closure of the non-generic strata). *Every null-collision stratum (degeneracy in Σ_N , resolved spectrum generic) carries an explicit complete invariant:*

(spec Σ_R , spec Σ_N with multiplicities, {generalized canonical correlations}, discrete gauge).

In the general case (simultaneous degeneracy in both sectors) the slice theorem gives the complete local model $O(q) \times_{O(m)} S$ and reduces the classification to the orbit space of the explicit slice representation of $\prod_i O(m_i)$ —a well-posed finite-dimensional orthogonal-orbit problem. The non-generic strata are thereby closed to the same standard as the generic stratum.

5 What is closed, and what is not

Closed. The local classification on the non-generic strata (Corollary 4.3): explicit and complete on the collision strata, reduced to explicit slice-orbit problems in general. This was the open frontier of the *local* theory, and it is now resolved by identifying the correct tool (the slice theorem) and computing the slice representations.

Non-claim 5.1 (The other two open pieces are separate). This closes the strata, not the framework. Two open pieces remain and are not strata questions: the global character variety $\text{Hom}(\pi_1, \text{Aut}(\mathcal{G}))/\text{conj}$, which is *bounded* (compact, explicit dimension, symplectic) but whose general computation is a hard problem of mathematics, not of this framework; and the irreducible external core of the null covariance, which is *provably irreducible* (one cannot recover directions never coupled to) and is therefore closed as an impossibility rather than filled. Neither is advanced or claimed here.

No physics. “Chaos” is spectral near-degeneracy. The diabolical phase is the monodromy of the null connection / slice representation; its identity with the physical Berry phase is formal. All tools are matrix perturbation theory, the slice theorem, and diabolical-point geometry. No physical claim is made.

6 Conclusion

The null sector is a stratified space whose orbit space is the closed Weyl chamber, whose faces are the multiplicity strata, and whose two degeneracy loci—the codimension-one coupled/free wall and the codimension-two conical intersection—are now understood in full. Null transport around a collision carries a $\mathbb{Z}/2$ diabolical phase, explained intrinsically as the monodromy of the weight-two slice representation; every boundary is as sharp as its spectral gap is wide. Above all, the non-generic strata—the last open frontier of the local theory—are closed: the Luna slice theorem supplies the complete local model at every stratum, the coupling is classified up to the residual orthogonal group by generalized canonical correlations, and the complete invariant extends explicitly to every collision stratum and, by reduction to slice orbits, to all of them. The local theory of resolution geometry is thereby complete on all strata, generic and non-generic alike; the coupling, which organized every earlier layer, organizes the degenerate ones too.

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